

Transtiff: Haptic Interaction with a Stick Interface with Various Stiffness

Ayumu Ogura
Aoyama Gakuin University
Sagamihara, Japan
ayumu.ogura@x-lab.team

Kodai Ito
Aoyama Gakuin University
Sagamihara, Japan
kodai.ito@it.aoyama.ac.jp

Shigeo Yoshida
OMRON SINIC X Corporation
Tokyo, Japan
shigeo.yoshida@sinicx.com

Kazutoshi Tanaka
OMRON SINIC X Corporation
Tokyo, Japan
kazutoshi.tanaka@sinicx.com

Yuichi Itoh
Aoyama Gakuin University
Sagamihara, Japan
itoh@it.aoyama.ac.jp

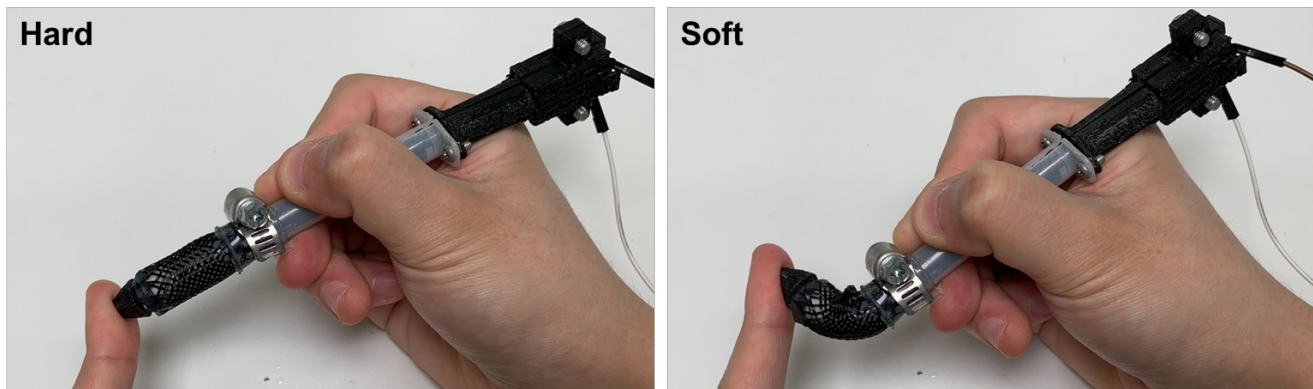


Figure 1: Transtiff is a stick-shaped device. The device is designed to augment the stick-based haptic experiences by controlling the stiffness of the device (left: hard; right: soft).

ABSTRACT

We propose *Transtiff*, a stick-shaped device that can display various stiffness for stick-based haptic interaction. The device has a stiffness-changing joint replicating an artificial muscle mechanism in the relay portion of the stick to change its stiffness. Transtiff can be applied to touch interaction of the screen, augmenting the haptic experience of operating with a stylus pen, which is usually felt uniform. As applications, users can experience the sensation of pen and brush writing on a single device. In addition, it is possible to change the stiffness of the device for each object on the screen to reproduce the tactile feel of that object.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction devices*; *Haptic devices*.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH '23 Emerging Technologies, August 06–10, 2023, Los Angeles, CA, USA
© 2023 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0154-2/23/08.
<https://doi.org/10.1145/3588037.3595402>

KEYWORDS

HCI (human computer interaction), haptic device, bending, pen interaction, fluid-based interface

ACM Reference Format:

Ayumu Ogura, Kodai Ito, Shigeo Yoshida, Kazutoshi Tanaka, and Yuichi Itoh. 2023. Transtiff: Haptic Interaction with a Stick Interface with Various Stiffness. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH '23 Emerging Technologies)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588037.3595402>

1 INTRODUCTION

We manipulate stick objects, such as chopsticks and pens, in our daily life. As the senses of the human hand are sensitive, we can recognize the characteristics of the object touching or manipulating through the stick. Then, we came up with an idea to augment the haptic experience of touching or manipulating something by controlling the property of the stick.

In this paper, we propose *Transtiff* (Figure 1), a stick-shaped device with joint stiffness control by artificial muscle mechanism. We focus on stiffness because the stick changes its overall behavior by only controlling the stiffness of its joints. We use a piston mechanism that uses a small motor to compress the liquid in a flexible tube to control the stiffness of the joint. We demonstrate its applications, writing experience in paint applications, and touch experience of on-screen objects.

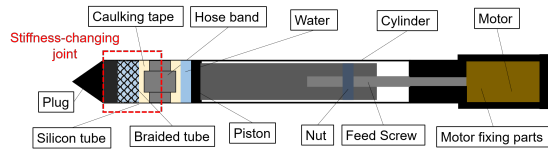


Figure 2: Structure of Transtiff.

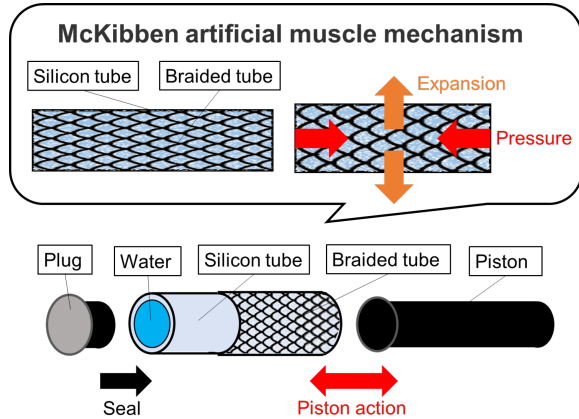


Figure 3: Mechanism of the stiffness change.

2 DESIGN AND IMPLEMENTATION

Figure 2 shows the structure of the Transtiff, and Figure 3 shows the mechanism of the stiffness change of the joint. The rotation of a feed screw connected to a motor (DC motor, 60 rpm) causes a nut to move linearly, enabling the reciprocating motion of an attached piston. The piston pushes the liquid in the cylinder toward the silicone tube. The tube section is equipped with a McKibben-type artificial muscle mechanism. The piston compresses water into the silicone tube, and the braided tube holds the tube from expanding. It causes the tube to expand in the radial direction, increasing the overall stiffness. We achieved the piston action by switching the direction of rotation of the motor. In the current prototype, we control the direction of rotation by operating scroll bars displayed on the GUI. To prevent liquid leakage, we inserted a plug-in shape of a pen nib on the tip side, wrapped caulking tape around the cylinder side, and tightened it with a hose band. The plug, piston, and motor mount were fabricated with a 3D printer. Arduino UNO is used to control the device.

To quantitatively confirm the change in stiffness of the Transtiff, we connected the tip of the fixed Transtiff to a force gauge and measured the maximum load (N) when the device was bent 10 mm. We took five measurements, each under two conditions: with and without piston pressurization, and calculated an average value. As a result, we obtained $0.865 \pm 4.150 \times 10^{-4}$ N for the no-pressure condition and $1.869 \pm 4.053 \times 10^{-3}$ N for the pressurized state. The force required to bend the tip of the device increased by a factor of 2.161 due to the pressurization of the piston.

3 APPLICATIONS

Transtiff provides a haptic sensation through dynamic changes of hard-soft feedback. Since stick-shaped objects we use in our daily

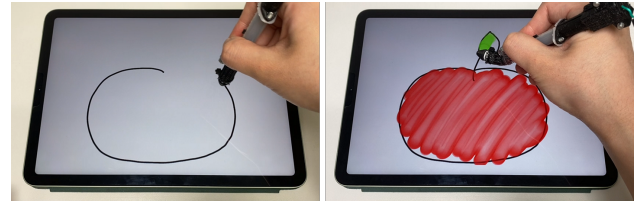


Figure 4: Paint application (left: pen outlining, right: brush coloring).



Figure 5: Haptic touch application (left: touching the diamond with hard sense, right: touching the marshmallow with soft sense).

lives have different stiffness, changing the stiffness of the Transtiff makes it possible to reproduce the feeling of using them. In addition, different haptic feedback when touching the same thing can present the sensation of touching another. Here, we introduce two possible applications that can be achieved by Transtiff.

First, we propose a painting application. In the actual painting experience, we do not use only one painting tool but different tools, such as pens and brushes, depending on what we want to paint. At this time, we feel other haptic sensations depending on the stiffness of the painting tools. In recent years, digital painting has become widespread, but existing stylus pens do not provide the soft haptic sensation like a brush. However, Transtiff enables a soft writing experience in addition to the traditional hard writing experience (Figure 4). It provides haptic experiences of various painting tools without changing devices in digital painting. Moreover, as Nicholas et al. [Fellion et al. 2017] have shown, flexible stylus pens positively impact specific input tasks. From this, we can find the usefulness of Transtiff.

Second, we propose a haptic touch application. Generally, touching the screen with a stylus provides only the same hard sensation. However, users can feel the tactile sense of each object on the screen by Transtiff (Figure 5). It allows the user to the hard touch of a diamond and the soft touch of a marshmallow. By adding the parameter “stiffness” to objects, we can experience more intuitive touch interaction on the screen.

ACKNOWLEDGMENTS

This work was partially supported by JSPS KAKENHI Grant Number JP22K18424.

REFERENCES

- Nicholas Fellion, Thomas Pietrzak, and Audrey Girouard. 2017. FlexStylus: Leveraging Bend Input for Pen Interaction. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology* (Québec City, QC, Canada) (UIST '17). Association for Computing Machinery, New York, NY, USA, 375–385. <https://doi.org/10.1145/3126594.3126597>